Gao et al. 2021 — Breast Cancer (10x) · CCCA cell-state re-analysis

Triple-negative breast carcinoma + DCIS, 10x scRNA-seq; 6,411 breast malignant cells (5 patients after threshold), scoped out of an 11-sample breast+thyroid bundle.

GO / NO-GO GATES

GATE 1: NO-GO ***(mixing + continuity)***

After Harmony on sample, a single connected kNN graph and iLISI 1.00→1.14 (of max 5) read as "integrated" — BUT the embedding is organised BY PATIENT: per-patient self-neighbour enrichment 2.3–5.6x at theta=10; the worst GROWS with theta (6.0/5.6/6.0/7.6x over theta 5/10/20/40, so Harmony over-corrects rather than merges) and is PC-count invariant (5.6x at n_pcs 20/30/40, ruling out under-powering). Every patient carries the same embryonic modules → identity, not biology. Clusters by sample: do not force a trajectory.

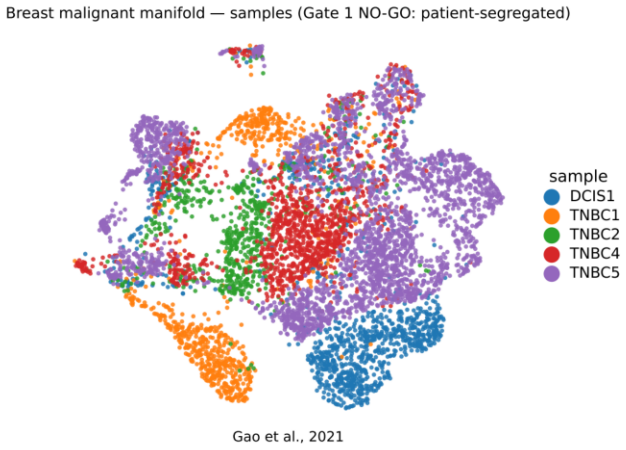
GATE 2: NA ***(module geography)***

Not evaluated — Gate 1 NO-GO halts the chain before the trajectory / signature steps (a valid, sample-integrated cell-state manifold is a precondition).

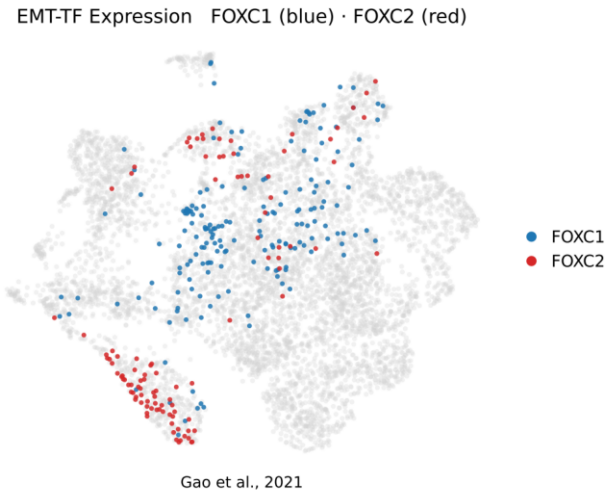
GATE 3: NA ***(hub cross-repression)***

Not evaluated — branching-only, and no trajectory exists.

STEP FIGURES



Gate 1 evidence: breast malignant manifold coloured by sample — patient-segregated



EMT-TF expression: FOXC1 (blue) / FOXC2 (red) thresholded overlay (descriptive)

KEY PARAMETER CHOICES + RATIONALE

Lineage scoping: The JoinKey bundles breast (TNBC1–5 + DCIS1) AND anaplastic thyroid carcinoma (ATC1–5). Scoped to breast (declared Category) on lineage markers: EPCAM/KRT8/18/19 high vs PAX8/TG~0.

Sample threshold: ≥347 cells (data-driven multiplicative gap 618→195, ratio 3.17); kept 5/6 breast samples → 6,411 cells (dropped TNBC3, 195 cells).

n_pcs / integration: n_pcs=20 (precedent-bounded: Kurten 20 / Venteicher 20 / Puram 40). PC1 = 31% of variance (an inter-patient / CNV axis). Harmony theta swept 5/10/20/40; theta=10 gives best mixing.

Geometry: None — Gate 1 NO-GO, no manifold fit. Global module means Gast −0.024 / pSom +0.016 / Crest −0.042 (reproduces the prior ledger row).

EMT-TFs (unconditional): FOXC1 (28.5% expressing) + FOXC2 (16.5%) — the two broadest detectable factors; widened panel incl. FOXC/PRRX, VIM excluded (a marker, not a TF). Descriptive overlay only.

Hub identity / r: NA — no hub and no cross-repression axis (branching-only).

VERDICT — the pattern does NOT reproduce. The breast malignant compartment is a per-patient archipelago (inter-patient clonal / CNV heterogeneity), not a sample-integrated cell-state continuum. Gate 1 NO-GO → no trajectory fit; downstream notebooks not produced.